INCH-POUND

MIL-C-11015/19E
14 May 2001
SUPERSEDING
MIL-C-11015/19D
28 September 1970

MILITARY SPECIFICATION SHEET

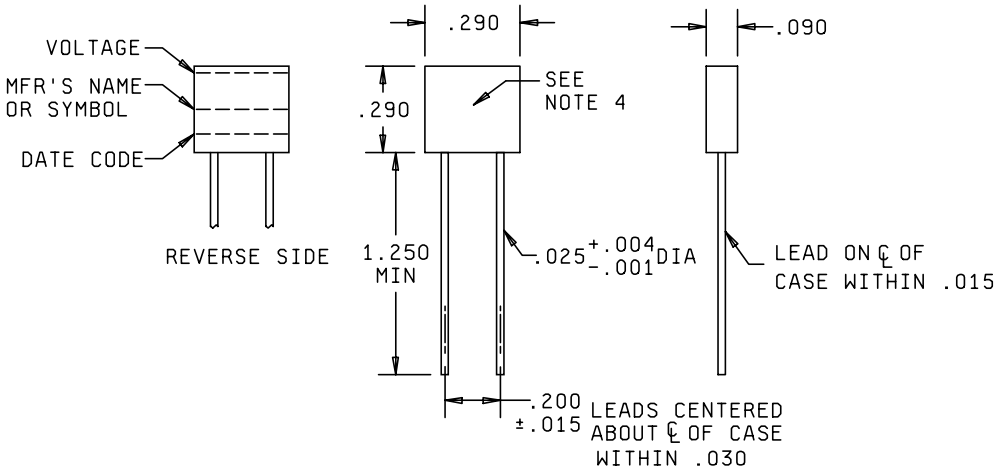
CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE),

STYLE CK06

The requirements for acquiring the capacitors described herein shall consist of this specification and MIL-C-11015.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN
after 28 September 1970. For new design use MIL-PRF-39014/2.




| Inches | Mm |
|---|---|
| .001 | .03 |
| .004 | .10 |
| .015 | .38 |
| .025 | .64 |
| .030 | .76 |
| .090 | 2.29 |
| .200 | 5.08 |
| .290 | 7.37 |
| 1.250 | 31.75 |

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (.25 mm).
3. Metric equivalents are given for general information only.
4. PIN marking to be divided into two or three lines; one line marking shall not be used.
5. Lead length may be a minimum of .625 inch (15.88 mm) for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Style CK06 capacitors.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type - Flat plate, molded or pre-formed.

Capacitance value - See table I.

Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.

Rated temperature - -55°C to +125°C
(although 1,200 pF through 10,000 pF capacitors are identified as B units, they shall have the capability of -55°C to +150°C operation, see figure 2.)

Dielectric withstanding voltage (DWV): In accordance with MIL-C-11015.

Dielectric:

Test voltage - 250 percent of rated voltage.

Body insulation:

Test potential - 1,300 volts dc.

Barometric pressure (reduced): In accordance with MIL-C-11015 and method 105 of MIL-STD-202, test condition D (100,000 ft) (0.315 inch (8.00 mm) of mercury).

Test potential - 150 percent of rated voltage.

Insulation resistance (IR): In accordance with MIL-C-11015; rated voltage applied.

100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.

Dissipation factor (DF): 2.5 percent, max.

Shock (specified pulse): In accordance with MIL-C-11015 and method 213 of MIL-STD-202, condition I (100 G's).

Vibration, high frequency: In accordance with MIL-C-11015 and method 204 of MIL-STD-202, condition D (20 g's).

Thermal shock and immersion: In accordance with MIL-C-11015.

IR - Not less than 50 percent of initial 25°C requirement.



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TABLE I. Style CK06 characteristics.

| PIN 1/ | Rated Voltage | Rated temperature and voltage-temperature limits | Capacitance | Capacitance tolerance |
|---|---|---|---|---|
| | Volts, dc | | pF | |
| CK06BX122K | 200 | BX | 1,200 | K |
| CK06BX152- | 200 | BX | 1,500 | K, M |
| CK06BX182K | 200 | BX | 1,800 | K |
| CK06BX222- | 200 | BX | 2,200 | K, M |
| CK06BX272K | 200 | BX | 2,700 | K |
| CK06BX332- | 200 | BX | 3,300 | K, M |
| CK06BX392K | 200 | BX | 3,900 | K |
| CK06BX472- | 200 | BX | 4,700 | K, M |
| CK06BX562K | 200 | BX | 5,600 | K |
| CK06BX682- | 200 | BX | 6,800 | K, M |
| CK06BX822K | 200 | BX | 8,200 | K |
| CK06BX103- | 200 | BX | 10,000 | K, M |
| CK06BX123K | 100 | BX | 12,000 | K |
| CK06BX153- | 100 | BX | 15,000 | K, M |
| CK06BX183K | 100 | BX | 18,000 | K |
| CK06BX223- | 100 | BX | 22,000 | K, M |
| CK06BX273K | 100 | BX | 27,000 | K |
| CK06BX333- | 100 | BX | 33,000 | K, M |
| CK06BX393K | 100 | BX | 39,000 | K |
| CK06BX473- | 100 | BX | 47,000 | K, M |
| CK06BX563K | 100 | BX | 56,000 | K |
| CK06BX683- | 100 | BX | 68,000 | K, M |
| CK06BX823K | 100 | BX | 82,000 | K |
| CK06BX104- | 100 | BX | 100,000 | K, M |
| CK06BX124K | 50 | BX | 120,000 | K |
| CK06BX154- | 50 | BX | 150,000 | K, M |
| CK06BX184K | 50 | BX | 180,000 | K |
| CK06BX224- | 50 | BX | 220,000 | K, M |
| CK06BX274K | 50 | BX | 270,000 | K |
| CK06BX334- | 50 | BX | 330,000 | K, M |
| CK06BX394K | 50 | BX | 390,000 | K |
| CK06BX474- | 50 | BX | 470,000 | K, M |
| CK06BX564K | 50 | BX | 560,000 | K |
| CK06BX684- | 50 | BX | 680,000 | K, M |
| CK06BX824K | 50 | BX | 820,000 | K |
| CK06BX105- | 50 | BX | 1,000,000 | K, M |

1/ Where applicable, the complete PIN will include an additional symbol to indicate capacitance tolerance.



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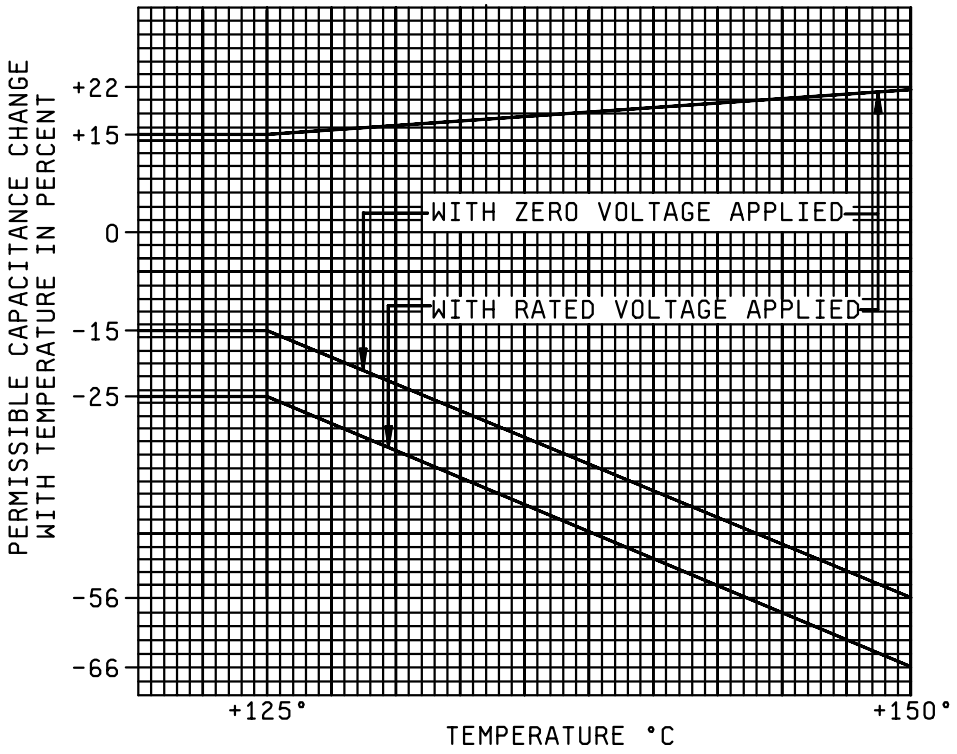


FIGURE 2. Extended voltage-temperature limits (1,200 pF through 10,000 pF capacitors).

Salt spray (corrosion): Not applicable.

Moisture resistance: In accordance with MIL-C-11015.

DWV - 250 percent of rated voltage.

IR - Not less than 10 percent of initial requirement.

Cap. - Within ±10 percent of initial measured value.

Solderability: In accordance with MIL-C-11015; 2 terminals.

Resistance to soldering heat: In accordance with MIL-C-11015.

IR - 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.

ΔCap. - +15, -5 percent of initial measurement.

DF - 2.5 percent, max.

Voltage-temperature limits: In accordance with MIL-C-11015 and figure 2 (for 1,200 pF through 10,000 pF capacitors).



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Life (at elevated ambient temperature): In accordance with MIL-C-11015.

Test potential - 200 percent of rated voltage.

IR - 1,000 megohms, min or 10 megohm-microfarads, min, whichever is less (at 125°C), and not less than 10 percent of initial requirement (at 25°C).

ΔC - Shall not exceed ±20 percent from initial measured value.

DF - 2.5 percent, max (at 25°C).

Marking: In accordance with MIL-C-11015, and as shown in figure 1.

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Supersession data: Items having a "CW" voltage-temperature limit formerly covered by Rev. A, and earlier issues, of this specification sheet are superseded as indicated in table II.

TABLE II. Cross-reference of substitute items.

| Superseded item 1/ | Recommended substitute item 1/ | Superseded item 1/ | Recommended Substitute Item 1/ |
|---|---|---|---|
| CK06CW122K | CK06BX122K | CK06CW392K | CK06BX392K |
| CK06CW152- | CK06BX152- | CK06CW472- | CK06BX472- |
| CK06CW182K | CK06BX182K | CK06CW562K | CK06BX562K |
| CK06CW222- | CK06BX222- | CK06CW682- | CK06BX682- |
| CK06CW272K | CK06BX272K | CK06CW822K | CK06BX822K |
| CK06CW332- | CK06BX332- | CK06CW103- | CK06BX103- |

1/ Where required, the PIN will include comparable capacitance tolerance symbol.

Inactivated styles: For new design units of MIL-PRF-39014/2 should be used as listed in table III. Although table III lists only the highest failure rate level, lower failure rate levels may be considered. For substitutability of the inactivated styles, established reliability capacitors in MIL-PRF-39014/2 of the same capacitance value, tolerance, rated voltage and temperature coefficient can be used regardless of the failure rate designation.



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TABLE III. Cross reference of inactivated styles.

| From inactivated MIL-C-11015/19 | To MIL-PRF-39014/2 M(1.0) M39014/02- | From inactivated MIL-C-11015/19 | To MIL-PRF-39014/2 M(1.0) M39014/02- |
|---|---|---|---|
| CK06BX122K | 1201 | CK06BX333- | 1223 |
| CK06BX152K | 1202 | CK06BX393K | 1224 |
| CK06BX152M | 1203 | CK06BX473- | 1225 |
| CK06BX182K | 1204 | CK06BX563K | 1226 |
| CK06BX222K | 1206 | CK06BX683- | 1227 |
| CK06BX222M | 1207 | CK06BX823K | 1229 |
| CK06BX272K | 1208 | CK06BX104- | 1230 |
| CK06BX332K | 1209 | CK06BX124K | 1233 |
| CK06BX332M | 1210 | CK06BX154- | 1234 |
| CK06BX392K | 1211 | CK06BX184K | 1235 |
| CK06BX472K | 1212 | CK06BX224- | 1236 |
| CK06BX472M | 1213 | CK06BX274K | 1237 |
| CK06BX562K | 1214 | CK06BX334- | 1238 |
| CK06BX682K | 1215 | CK06BX394K | 1239 |
| CK06BX682M | 1216 | CK06BX474- | 1240 |
| CK06BX822K | 1217 | CK06BX564K | 1404 |
| CK06BX103K | 1218 | CK06BX684- | 1405 |
| CK06BX103M | 1219 | CK06BX824K | 1406 |
| CK06BX123K | 1231 | CK06BX105- | 1407 |
| CK06BX153- | 1220 | | |
| CK06BX183K | 1221 | | |
| CK06BX223- | 1222 | | |
| CK06BX273K | 1232 | | |

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA – CC

Review activities:
Army – AR, EA, MI
Navy - MC
Air Force - 19

Preparing activity:
DLA - CC

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